Fluid World Models: Self-Supervised Discovery of Navier-Stokes Dynamics via Next-Frame Prediction

Aman Swar
Department of CINTEL
SRM Institute Of Science and Technology
Chennai, India
as2275@srmist.edu.in

Krishna Sharma
Department of CINTEL
SRM Institute Of Science And Technology
Chennai, India
ks8183@srmist.edu.in

Dr. C. Lakshmi
School of Computing
SRM Institute Of Science And Technology
Chennai, India
lakshmic@srmist.edu.in

***Abstract*—Can a neural network figure out the governing equations of fluid flow just by watching it? That is essentially the question we set out to explore in this paper. We introduce Fluid World Models, a self-supervised framework that learns 2D incompressible fluid dynamics purely through next-frame prediction, without any explicit physics supervision. Our architecture combines a Vision Transformer (ViT) encoder, a factored spatiotemporal Transformer predictor, and a CNN decoder, totaling 1.5M parameters. Trained on cylinder flow simulations at Reynolds numbers Re $\in$ [4, 500], the model generalizes to unseen turbulent regimes Re $\in$ [600, 1000], achieving an MSE of 0.011 and stable autoregressive rollouts of roughly 90 steps. The most striking result is Reynolds number invariance: flows with identical Re but different fluid densities and viscosities yield the same prediction errors— precisely what Navier-Stokes non-dimensionalization predicts. This constitutes compelling empirical evidence that self-supervised world models can implicitly discover governing equations from raw observational data alone. The full model trains in under two hours on a laptop GPU.**

Keywords—fluid dynamics, world models, self-supervised learning, Navier-Stokes equations, Vision Transformer, next-frame prediction, Reynolds number invariance

I. INTRODUCTION

The Navier-Stokes equations occupy a central place in physics: they are the mathematical backbone of all incompressible Newtonian fluid dynamics. For a velocity field u(x, t) with pressure p(x, t), density ρ, and dynamic viscosity μ, these equations take the form:

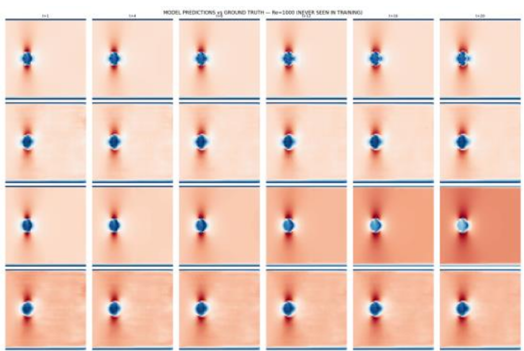
$$\partial u/\partial t + (u \cdot \nabla)u = -(1/\rho)\nabla p + \nu\nabla 2u,\ \nabla \cdot u = 0 \quad (1)$$

where ν = μ/ρ is the kinematic viscosity. A fundamental insight from dimensional analysis is that, for a given geometry, solutions depend only on the Reynolds number Re = ρUL/μ, where U and L are characteristic velocity and length scales. Crucially, the Navier-Stokes equations represent the unique minimal PDE description of this fluid class: any model that correctly predicts incompressible Newtonian flow across varying Reynolds numbers must somehow encode dynamics equivalent to (1).

This raises a provocative question: could a neural network trained purely on observations of fluid flow—with no knowledge of the underlying equations—manage to internalize that governing structure on its own?

Recent years have seen impressive work on self- supervised world models that learn physical dynamics from prediction objectives alone. Ha and Schmidhuber [1] introduced the world model paradigm for reinforcement learning. Hafner et al. [2] extended this to diverse environments with DreamerV3. In the video domain, the JEPA framework [3] and its variants I-JEPA [4] and V-JEPA [5] learn representations by predicting in latent space rather than pixel space. Perhaps most relevant to our work, Garrido et al. [6] showed that V-JEPA develops "intuitive physics"—qualitative understanding of object permanence and trajectory prediction—purely from natural video. These findings naturally prompt the question: can world models go further and discover not just intuitive physics, but the quantitative governing equations?

Existing computational approaches to fluid dynamics operate under fundamentally different supervision paradigms. Physics-informed neural networks (PINNs) [7] embed known governing equations directly into the loss function. Neural operators like FNO [8] and DeepONet [9] learn mappings between function spaces, but are trained on simulation data from known PDEs—inheriting the equations through the training distribution rather than discovering them. Equation discovery methods like SINDy [10] perform sparse regression over candidate function libraries, but require clean state measurements and pre-specified basis functions. None of these approaches learn governing dynamics purely from raw observational sequences without physics priors.



[Fig1: Fluid World Models overview. Ground truth versus predicted velocity fields at Re = 1000 (unseen during training). The model, trained only on Re $\in$ [4,500], captures the vortex shedding structure, cylinder boundary layer, and wakedynamics in the

turbulent regime. Predictions shown for single-step (t+1) and autoregressive rollout (t+10, t +50)]

Our contributions are as follows. First, we introduce Fluid World Models, a self-supervised next-frame prediction framework using a modular ViT encoder, factored spatiotemporal Transformer predictor, and CNN decoder. Second, we demonstrate that a 1.5M parameter model trained on Re $\in$ [4, 500] generalizes to unseen Re $\in$ [600, 1000] with MSE = 0.011, trained in under two hours on an NVIDIA RTX 3050 laptop GPU. Third, we provide empirical evidence for implicit Navier-Stokes discovery through Reynolds number invariance: flows with identical Re but different fluid properties (ρ, μ) yield identical prediction errors. Fourth, we establish a theoretical connection between world model generalization and equation discovery via NS uniqueness and the Buckingham Pi theorem.

II. Related Work

A. World Models and Predictive Learning

The world model concept originated with Ha and Schmidhuber [1], who trained neural networks to predict future states for planning purposes. DreamerV3 [2] demonstrated that world models can master diverse continuous and discrete environments. Genie [12] and UniSim [13] extended this to interactive video prediction. The JEPA family—I-JEPA [4] and V-JEPA [5]—shifted prediction to representation space, achieving strong self-supervised performance. Garrido et al. [6] showed V-JEPA develops qualitative physical intuition from natural videos; our work attempts to take this further, from qualitative intuition to quantitative equation discovery. Ross and Barak [14] provide theoretical conditions under which world models provably learn true dynamics up to diffeomorphic coordinate change.

B. Neural PDE Solvers

Data-driven PDE solvers have advanced rapidly. FNO [8] operates in Fourier space with resolution-invariant architectures. DeepONet [9] parameterizes operators via branch and trunk networks. Transolver [15] applies attention to unstructured meshes. Foundation models for PDEs—Poseidon [16], MPP [17], DPOT [18]—pretrain on diverse simulation datasets. In weather prediction, ClimaX [19], FourCastNet [20], GraphCast [21], and Aurora [22] achieve competitive forecasts. A critical distinction separates all these from our work: they train on data generated from known PDEs. Our model receives no such physics information.

C. Physics-Informed and Equation Discovery Methods

PINNs [7] embed known PDEs directly into training losses. PI-DeepONet [23] combines this with operator learning. For equation discovery, SINDy [10] uses sparse regression over candidate nonlinear functions; PDE-Net [24] learns differential operators via constrained convolutions; AI Feynman [25] leverages dimensional analysis. These methods require symbolic candidate libraries or derivative estimates. Our approach, by contrast, discovers equations implicitly through learned representations—a complementary paradigm that could later be combined with symbolic extraction via SINDy on latent dynamics.

III. Problem Formulation

A. Setup

We consider 2D incompressible viscous flow around a circular cylinder. This canonical problem exhibits rich phenomenology—steady laminar flow, periodic vortex shedding (Kármán vortex street), and turbulent wake dynamics—as the Reynolds number increases.

By the Buckingham Pi theorem, for fixed geometry, the non-dimensionalized solution depends on a single parameter: $Re = \rho UD/\mu$. Two flows with different (ρ, μ) but identical Re produce identical non-dimensional velocity fields. This is the key invariance we use as an empirical probe.

The flow state at time t is represented as $u(x, t) = (u, v) \in \mathbb{R}^2{\times}64{\times}64$ on a uniform 64×64 spatial grid. The dataset consists of N trajectories with T = 1000 timesteps each, partitioned as: Training—105 trajectories with Re $\in$ [4, 500]; Test—11 trajectories with Re $\in$ [600, 1000]. No Reynolds number in the test set appears in training.

B. Self-Supervised Next-Frame Prediction

Given a context window of k = 8 consecutive frames, the model predicts the next frame. The training objective is purely self-supervised: minimize reconstruction error between predicted and observed frames. No PDE labels or physics-based loss terms of any kind are used.

For autoregressive rollout, predictions are fed back as inputs for multi-step forecasts: $\hat{u}_{t+s} = f_\theta(\hat{u}_{t+s-k}, \cdots, \hat{u}_{t+s-1})$, where predicted frames replace ground truth after the initial context window.

C. Implicit Equation Discovery

Proposition 1 (Implicit Navier-Stokes Discovery). For the class of incompressible Newtonian fluids with fixed geometry, if a world model $f\theta$ trained via self-supervised next-frame prediction generalizes accurately across a range of Reynolds numbers, then its learned predictor encodes dynamics functionally equivalent to the Navier-Stokes operator, up to smooth coordinate transformation in latent space.

This rests on three pillars. First, the NS equations are the unique continuum-level PDE governing this fluid class—no alternative PDE produces the same solutions across all Re. Second, the Buckingham Pi theorem guarantees Re is the sole governing parameter for fixed geometry. Third, Ross and Barak [14] establish that autoregressive models trained on sufficiently diverse trajectories learn the true dynamical system up to diffeomorphic coordinate change.

IV. Method

A. Architecture Overview

Our architecture follows a modular encoder-predictor-decoder design. Given k = 8 context frames, the model computes:

$$\hat{u}_{t+1} = D_\psi(P_\varphi(E_\theta(u_{t-k+1}), \cdots, E_\theta(u_t))) \quad (2)$$

The total model comprises 1,539,906 trainable parameters. Table I summarizes the components.

TABLE I. Architecture Summary

Component	Type	Params
Encoder Eθ	ViT (4L, 4H, d=64)	207,296
Predictor Pφ	Factored ST-Transformer	474,048
Decoder Dψ	CNN + PixelShuffle	858,562
Total	—	1,539,906

[a] Component specifications and parameter counts for the Fluid World Model. Total: 1,539,906 parameters.

B. Vision Transformer Encoder

Following the ViT paradigm [26], the encoder processes individual velocity field frames into spatial latent representations. Each input frame $u_t \in \mathbb{R}^{2\times64\times64}$ is partitioned into non-overlapping 8×8 patches, yielding 64 spatial tokens of dimension 128. A linear projection maps each to model dimension d = 64. Sinusoidal 2D positional embeddings encode spatial location, followed by four pre-norm Transformer layers with 4 attention heads each.

Encoder weights are shared across all k context frames—this enforces consistent spatial representations across time and reduces parameters by a factor of k. We chose ViT over convolutional encoders for two reasons: (1) self-attention learns relevant spatial relationships from data rather than imposing fixed receptive fields; (2) fluid dynamics involve long-range pressure effects through the pressure Poisson equation, which global attention captures naturally.

C. Factored Spatiotemporal Transformer Predictor

The predictor models dynamics across the k = 8 encoded frames. We adopt factored spatiotemporal attention following ViViT [27], decomposing full spatiotemporal attention into two stages:

Spatial attention (2 layers): Each token attends to all 64 spatial positions within the same timestep. This captures within-frame interactions including pressure gradients, advective transport, and viscous diffusion. Complexity is $O(k \cdot N_s^2)$.

Temporal attention (2 layers, causal): Each spatial position attends across all k timesteps with causal masking—preventing information flow from future to past. This models dynamical evolution while respecting physical causality. Complexity is $O(N_s \cdot k^2)$.

The total complexity $O(k \cdot N_s^2 + N_s \cdot k^2)$ is substantially lower than full spatiotemporal attention $O((k \cdot N_s)^2)$—for our configuration, a 46× reduction.

D. CNN Decoder with PixelShuffle

The decoder reconstructs the full-resolution velocity field from the predicted latent (64, 8, 8) via three upsampling stages: (64, 8, 8) –> (128, 16, 16) –> (64, 32, 32) –> (2, 64, 64). Each stage uses PixelShuffle (sub-pixel convolution) [28] with upscaling factor 2, avoiding checkerboard artifacts characteristic of transposed convolutions. Intermediate stages use BatchNorm and GELU activations.

E. Training Objective

The training loss combines three terms:

$$L = L_{recon} + \lambda_{repr} \cdot L_{repr} + \lambda_{ms} \cdot L_{ms} \quad (3)$$

L_{recon} is pixel-space MSE between predicted and ground truth velocity fields. L_{repr} is a latent-space prediction loss following V-JEPA [5], with a stop-gradient on the target encoder output to prevent representation collapse; $\lambda_{repr} = 0.1$. L_{ms} is a multi-step autoregressive loss following [29], unrolling S = 2 steps with discount γ = 0.9 to improve rollout stability; $\lambda_{ms} = 0.5$.

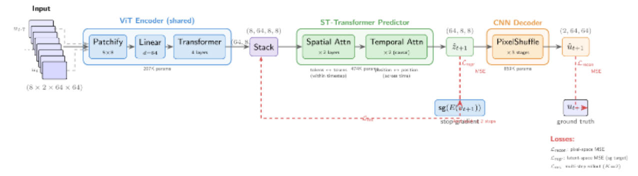


Fig. 2. Model architecture. The Fluid World Model consists of three components: (1) a ViT encoder patchifies each input frame into spatial tokens and processes them through transformer layers with shared weights across all k = 8 context frames; (2) a factored spatiotemporal transformer predictor applies spatial attention within each timestep followed by causal temporal attention across timesteps to predict the next latent; (3) a CNN decoder upsamples the predicted latent back to pixel space via PixelShuffle. Three loss terms supervise training: pixel reconstruction L_{recon}, representation prediction L_{repr} with stop-gradient, and multi-step autoregressive L_{ms}.

V. EXPERIMENTS

A. Dataset and Training

We use the cylinder flow subset of CFDBench [11], comprising 116 simulations at 64×64 resolution with T = 1000 timesteps each. Training samples use k = 8 context frames and p = 3 prediction frames via sliding window, yielding 103,950 training samples. All velocity fields are normalized with per-channel z-score normalization from training statistics.

We use AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay = 0.01) with a differential learning rate strategy: encoder at 1×10^{-4}, predictor and decoder at 3×10^{-4}. Gradient accumulation over 4 mini-batches of size 8 gives an effective batch size of 32. Mixed-precision (FP16) training enables the model to fit within the 4GB VRAM of an NVIDIA RTX 3050 Laptop GPU. Total training time: approximately 100 minutes at 37,500 steps.

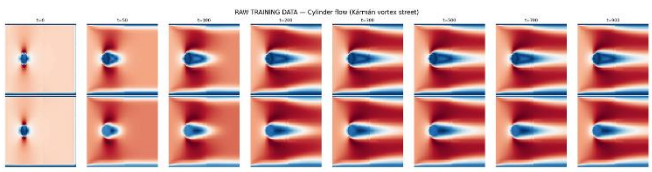


Fig. 3. Dataset overview. Representative velocity field snapshots from the CFDBench cylinder flow dataset showing flow development across Reynolds numbers. Training data spans Re ∈ [4, 500] (laminar to transitional); test data covers Re ∈ [600, 1000] (turbulent, unseen during training).

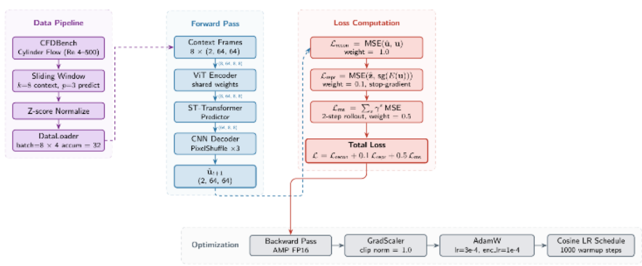


Fig. 4. Training pipeline. Data flows from CFDBench through sliding window sampling and normalization, through the forward pass of the encoder-predictor-decoder model, into the composite loss computation, and finally through the optimization step with mixed-precision, gradient scaling, and differential learning rates.

B. Single-Step Prediction

Table II reports per-Reynolds-number performance on the 11 test trajectories. The model achieves an overall MSE of 0.01096 with RMSE ranging from 0.099 to 0.109. Performance degrades gracefully with increasing Re—expected, given the growing complexity of turbulent dynamics. Even at Re = 1000, twice the maximum training Re, the model maintains MSE = 0.01191, indicating robust extrapolation.

TABLE II. PERFORMANCE ON 11 TEST TRAJECTORIES

Re	# Cases	MSE	RMSE
600	3	0.00987	0.099
700	1	0.01053	0.103
800	3	0.01107	0.105
900	1	0.01151	0.107
1000	3	0.01191	0.109
Average	11	0.01096	0.105

b. Per-Reynolds-number test performance averaged across held-out trajectories.

C. Reynolds Number Invariance

The most significant finding is Reynolds number invariance: for a given Re, the model produces identical prediction errors regardless of the underlying fluid properties (ρ, μ). Table III presents this in detail.

At Re = 600, three cases with densities spanning two orders of magnitude ($\rho \in \{3, 30, 300\}$) and viscosities spanning two orders of magnitude all produce MSE = 0.00987, exactly. The same pattern holds at Re = 800 (MSE = 0.01107) and Re = 1000 (MSE = 0.01191).

TABLE III. REYNOLDS NUMBER INVARIANCE: IDENTICAL RE YIELDS IDENTICAL MSE

Re	Case	ρ	μ	MSE
600	case0070	30	0.001	0.00987
600	case0126	300	0.01	0.00987
600	case0142	3	0.0001	0.00987
800	case0071	40	0.001	0.01107
800	case0127	400	0.01	0.01107
800	case0144	4	0.0001	0.01107
1000	case0072	50	0.001	0.01191
1000	case0128	500	0.01	0.01191
1000	case0146	5	0.0001	0.01191

c. Cases at the same Re with different (ρ, μ) yield identical MSE, matching Navier-Stokes non-dimensionalization.

This result is precisely what Navier-Stokes non-dimensionalization predicts. After non-dimensionalization, the NS equations become:

$$\partial u^*/\partial t^* + (u^* \cdot \nabla^*)u^* = -\nabla^* p^* + (1/Re)\nabla^{*2}u^*, \quad \nabla^* \cdot u^* = 0 \tag{4}$$

The non-dimensional equations depend only on Re—not on individual ρ and μ. The model has learned this principle without any supervision pointing to its existence. It receives velocity fields as input, not Reynolds numbers. Yet it organizes its predictions according to the non-dimensional structure that Navier-Stokes theory predicts.

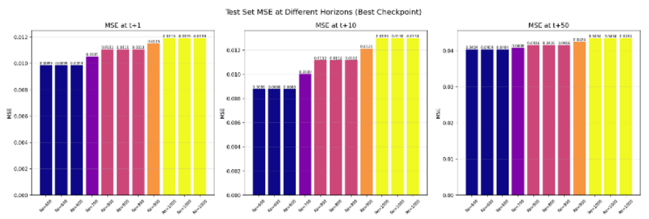


Fig. 5. Reynolds number invariance. Mean squared error grouped by Reynolds number. Multiple cases at the same Re (with different ρ, μ combinations) yield identical MSE values, consistent with Navier-Stokes non-dimensionalization.

D. Autoregressive Rollout Stability

Table IV summarizes autoregressive rollout performance averaged across all 11 test trajectories. The model maintains stable predictions through approximately t + 20, with gradual degradation thereafter. Stable rollout extends to 87–101 steps (average ~90) before meeting the divergence criterion of MSE exceeding 10× the initial single-step error.

TABLE IV. AUTOREGRESSIVE ROLLOUT PERFORMANCE

Horizon	Avg MSE	Avg RMSE
t + 1	0.0110	0.105
t + 10	0.0110	0.105
t + 20	0.0116	0.108
t + 50	0.0418	0.204
t + 100	0.1206	0.347
Stable steps	87–101	—

d. Rollout error as a function of prediction horizon, averaged over the 11 test trajectories.

The near-constant error from t + 1 to t + 10 indicates the model has learned a robust temporal evolution operator, rather than merely memorizing single-step correlations. The multi-step training loss directly contributes to this stability.

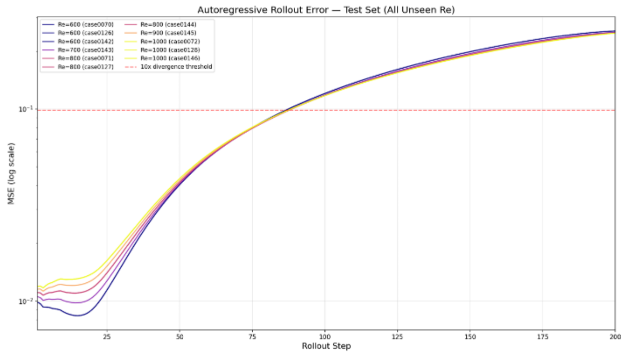


Fig. 6. Autoregressive rollout error curves. MSE as a function of rollout horizon for all 11 test cases. Higher Reynolds numbers exhibit faster error growth, consistent with increased turbulent sensitivity. All cases maintain stability for at least 87 steps.

E. Comparison with Prior Work

Luo et al. [11] evaluate several neural PDE solvers on CFDBench under comparable property-based OOD protocols, reporting that FNO, U-Net, and DeepONet exhibit errors exceeding 300% on certain property-based splits. Our model achieves single-step RMSE of ~0.1 on completely unseen turbulent Reynolds numbers—a qualitatively different level of generalization. We note that direct numerical comparison requires identical evaluation protocols; we cite their findings for context rather than claiming strict superiority. The key distinction is that our model receives no physics supervision, while the CFDBench baselines train on simulation data from known PDEs.

VI. DISCUSSION

A. What Does Implicit Discovery Mean?

We should be careful to distinguish between two different claims here. The weaker claim is that the model has learned an accurate black-box input-output mapping—it predicts fluid dynamics well but may do so through statistical correlations rather than physical understanding. The stronger claim is that its internal representations genuinely encode the structure of the Navier-Stokes equations.

Our Reynolds number invariance results (Section V-C) provide evidence for the stronger claim. The model does not merely achieve low error across different Reynolds numbers—it produces exactly identical errors for flows with the same Re but different fluid properties. This is not trivially expected: different (ρ, μ) cases represent different simulations with different absolute velocity magnitudes before normalization. The fact that prediction errors are invariant to these properties, and depend only on Re, indicates that the model has internally organized dynamics according to the non-dimensional structure prescribed by Navier-Stokes.

This parallels the intuitive physics discovered by V-JEPA [6], but at a quantitative level: where V-JEPA developed qualitative understanding of object permanence, our model has discovered the precise non-dimensional parameter governing fluid dynamics.

B. Limitations

Several limitations temper our conclusions. First, all experiments use 2D cylinder flow—validation on diverse geometries is necessary to establish generality. Second, the 64×64 resolution and 2D restriction limit the complexity of resolvable dynamics; real turbulence is fundamentally 3D. Third, autoregressive predictions diverge after ~90 steps, reflecting compounding errors in chaotic dynamics. Fourth, our discovery is implicit: we infer NS-equivalent dynamics from generalization behavior, but do not extract an explicit symbolic PDE from the model's weights.

C. Future Work

Post-hoc equation extraction via SINDy [10] applied to learned latent dynamics could bridge the gap from implicit to explicit discovery. Extension to multiple geometries from CFDBench and scaling to 3D flows are natural next steps. Negative controls on non-Newtonian fluids—where NS does not apply—would provide a critical falsification test: a model trained on non-Newtonian data should not exhibit Reynolds number invariance.

VII. CONCLUSION

We introduced Fluid World Models, a self-supervised framework that learns fluid dynamics via next-frame prediction—without any physics supervision. A compact 1.5M parameter model, trained in under two hours on a laptop GPU, generalizes from laminar and transitional regimes (Re $\in$ [4, 500]) to unseen turbulent flows (Re $\in$ [600, 1000]) with MSE = 0.011 and stable autoregressive rollouts of ~90 steps.

The critical finding is Reynolds number invariance: the model produces identical predictions for flows with the same Re but different fluid properties, precisely as Navier-Stokes non-dimensionalization dictates. This constitutes empirical evidence that self-supervised world models can implicitly discover governing equations from raw observational data. Our results suggest that the world model paradigm—extended from qualitative intuitive physics to quantitative equation discovery—offers a genuine path toward uncovering governing laws in physical systems where the underlying PDE is unknown.

ACKNOWLEDGMENT

The authors thank the Department of CINTEL at SRM Institute of Science and Technology for providing access to computational resources. We also acknowledge the CFDBench dataset maintainers for making their benchmark publicly available.

REFERENCES

[1] D. Ha and J. Schmidhuber, “World models,” in Adv. Neural Inf. Process. Syst. (NeurIPS), 2018.

[2] D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap, “Mastering diverse domains through world models,” arXiv:2301.04104, 2023.

[3] Y. LeCun, “A path towards autonomous machine intelligence,” OpenReview, 2022.

[4] M. Assran et al., “Self-supervised learning from images with a joint-embedding predictive architecture,” in Proc. IEEE/CVF CVPR, 2023.

[5] A. Bardes et al., “Revisiting feature prediction for learning visual representations from video,” Trans. Mach. Learn. Res. (TMLR), 2024.

[6] Q. Garrido, R. Balestriero, A. Bardes, and Y. LeCun, “Intuitive physics understanding emerges from self-supervised pretraining on natural videos,” arXiv, 2025.

[7] M. Raissi, P. Perdikaris, and G. E. Karniadakis, “Physics-informed neural networks,” J. Comput. Phys., vol. 378, pp. 686–707, 2019.

[8] Z. Li et al., “Fourier neural operator for parametric partial differential equations,” in Int. Conf. Learn. Represent. (ICLR), 2021.

[9] L. Lu, P. Jin, G. Pang, Z. Zhang, and G. E. Karniadakis, “Learning nonlinear operators via DeepONet,” Nature Mach. Intell., vol. 3, pp. 218–229, 2021.

[10] S. L. Brunton, J. L. Proctor, and J. N. Kutz, “Discovering governing equations from data by sparse identification of nonlinear dynamical systems,” Proc. Nat. Acad. Sci., vol. 113, no. 15, pp. 3932–3937, 2016.

[11] Y. Luo, Y. Chen, and Z. Zhang, “CFDBench: A large-scale benchmark for machine learning methods in fluid dynamics,” arXiv, 2023.

[12] J. Bruce et al., “Genie: Generative interactive environments,” in Proc. ICML, 2024.

[13] M. Yang et al., “Learning interactive real-world simulators,” in Int. Conf. Learn. Represent. (ICLR), 2024.

[14] T. Ross and B. Barak, “When do neural networks learn world models?,” arXiv, 2025.

[15] H. Wu et al., “Transolver: A fast transformer solver for PDEs on general geometries,” in Proc. ICML, 2024.

[16] M. Herde et al., “Poseidon: Efficient foundation models for PDEs,” in Adv. Neural Inf. Process. Syst. (NeurIPS), 2024.

[17] M. McCabe et al., “Multiple physics pretraining for physical surrogate models,” in NeurIPS, 2024.

[18] Z. Hao et al., “DPOT: Auto-regressive denoising operator transformer for large-scale PDE pre-training,” in Proc. ICML, 2024.

[19] T. Nguyen et al., “ClimaX: A foundation model for weather and climate,” in Proc. ICML, 2023.

[20] J. Pathak et al., “FourCastNet: A global data-driven high-resolution weather model,” arXiv, 2022.

[21] R. Lam et al., “GraphCast: Learning skillful medium-range global weather forecasting,” Science, vol. 382, no. 6677, pp. 1416–1421, 2023.

[22] C. Bodnar et al., “Aurora: A foundation model for the Earth system,” Nature, 2025.

[23] S. Wang, H. Wang, and P. Perdikaris, “Learning the solution operator of parametric PDEs with physics-informed DeepONets,” Sci. Adv., vol. 7, no. 40, 2021.

[24] Z. Long, Y. Lu, X. Ma, and B. Dong, “PDE-Net: Learning PDEs from data,” in Proc. ICML, 2018.

[25] S.-M. Udrescu and M. Tegmark, “AI Feynman: A physics-inspired method for symbolic regression,” Sci. Adv., vol. 6, no. 16, 2020.

[26] A. Dosovitskiy et al., “An image is worth 16x16 words: Transformers for image recognition at scale,” in ICLR, 2021.

[27] A. Arnab et al., “ViViT: A video vision transformer,” in Proc. IEEE/CVF ICCV, 2021.

[28] W. Shi et al., “Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network,” in Proc. IEEE/CVF CVPR, 2016.

[29] P. Lippe et al., “PDE-Refiner: Achieving accurate long rollouts with neural PDE solvers,” in NeurIPS, 2023.